AI ASSISTED CODING - 19

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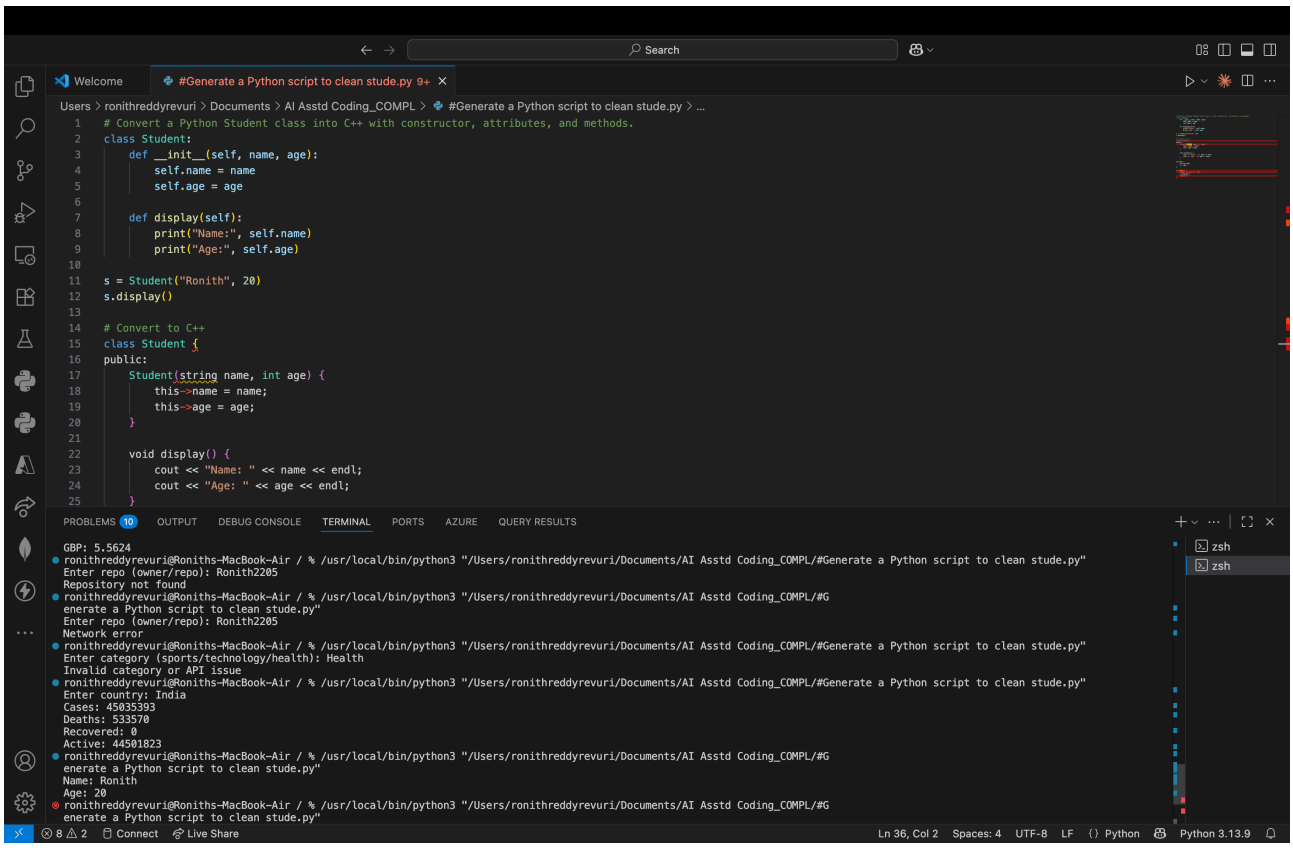
TASK 1 – PYTHON TO C++ CONVERSION

PROMPT :
CONVERT A PYTHON STUDENT CLASS INTO C++ WITH CONSTRUCTOR, ATTRIBUTES, AND METHODS.

EXPLANATION :

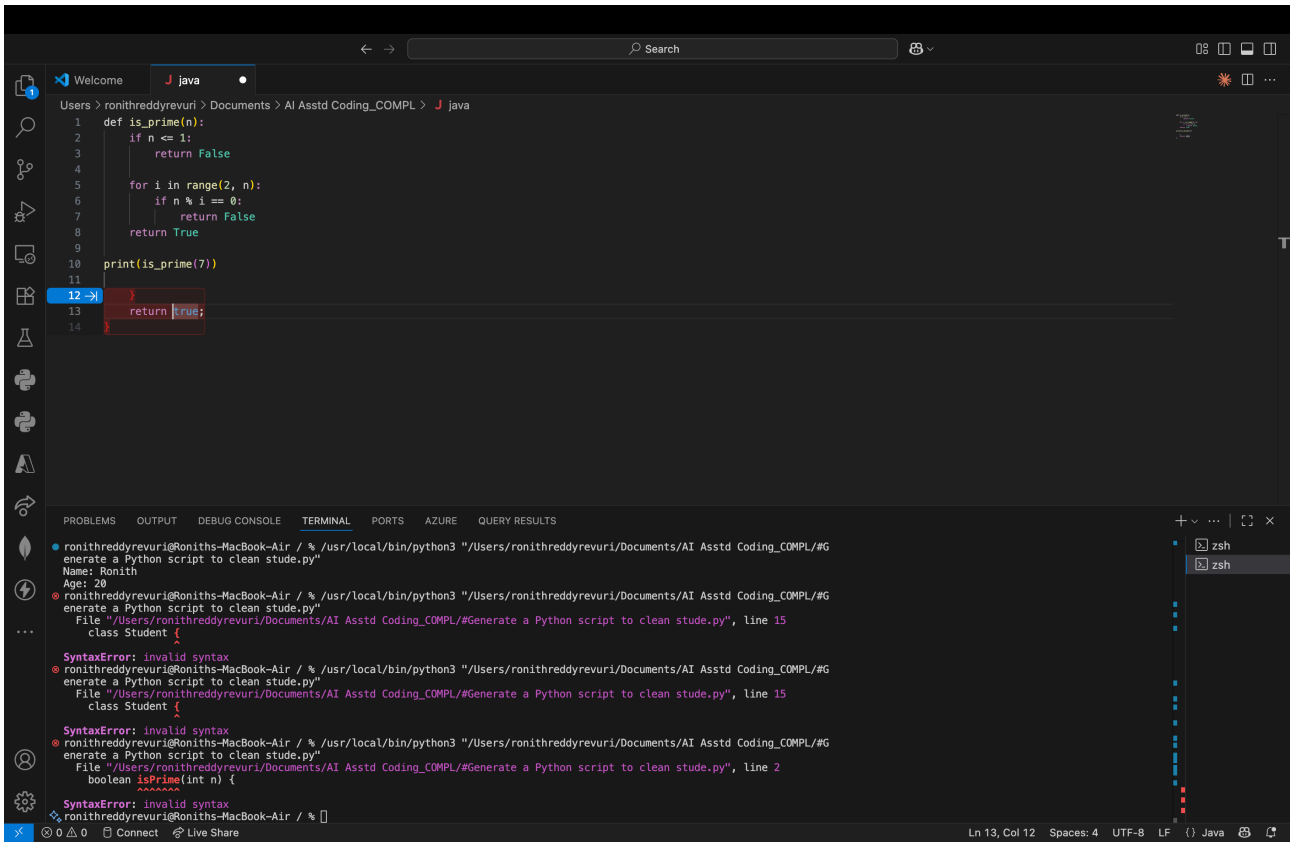
THE PYTHON CLASS IS CONVERTED INTO C++ USING PROPER DATA TYPES AND ACCESS SPECIFIERS. A CONSTRUCTOR IS ADDED TO INITIALIZE VALUES. BOTH VERSIONS GIVE THE SAME OUTPUT.



TASK 2 – JAVA TO PYTHON FUNCTION CONVERSION

PROMPT :
CONVERT JAVA ISPRIME METHOD INTO PYTHON FUNCTION.

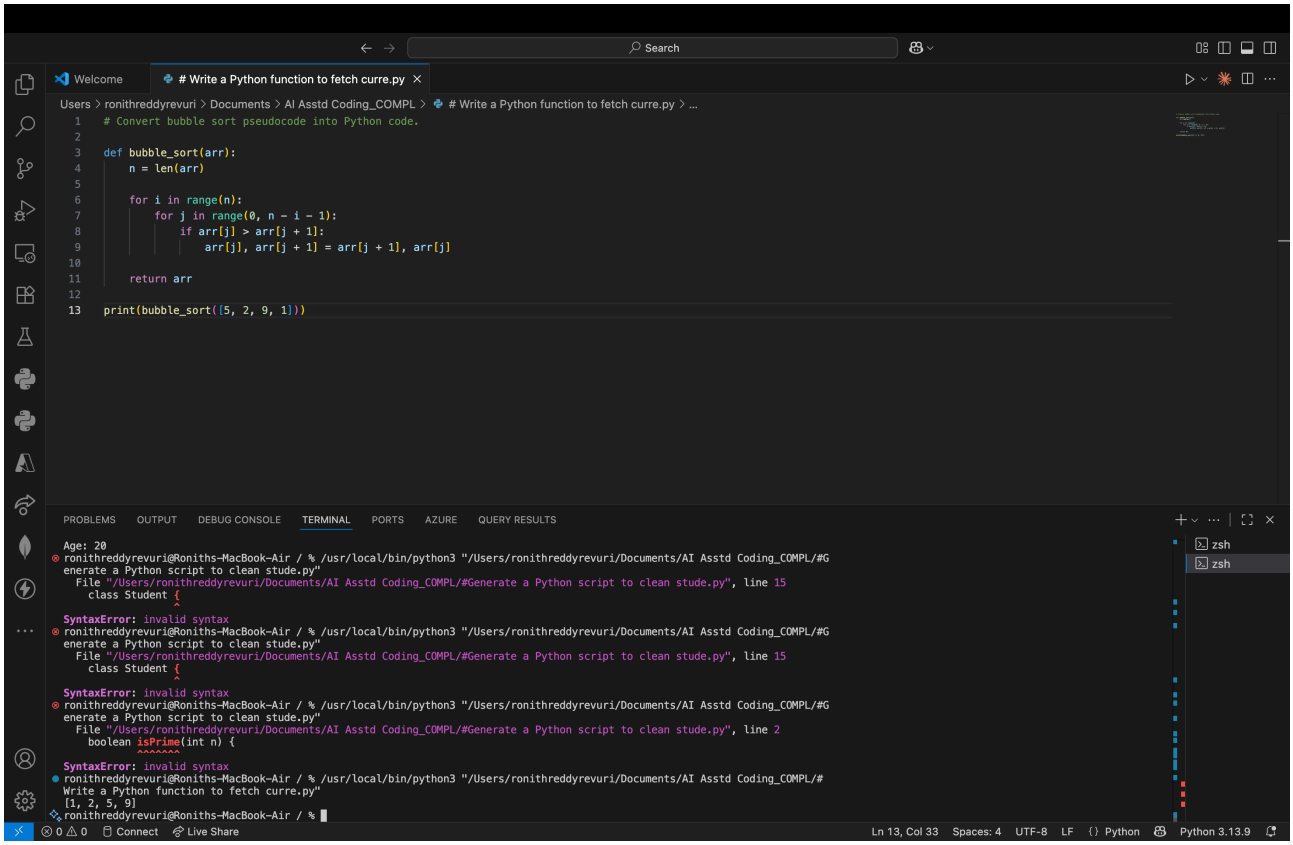
EXPLANATION :
THE JAVA METHOD IS REWRITTEN IN PYTHON USING SIMPLE SYNTAX. LOOPS AND CONDITIONS REMAIN THE SAME BUT PYTHON REMOVES SEMICOLONS AND USES INDENTATION.



TASK 3 – PSEUDOCODE TO PYTHON IMPLEMENTATION

PROMPT :
ASK 3 – PSEUDOCODE TO PYTHON IMPLEMENTATION

EXPLANATION :
THE PSEUDOCODE LOGIC IS IMPLEMENTED USING LOOPS IN PYTHON. ELEMENTS ARE COMPARED AND SWAPPED STEP BY STEP UNTIL THE LIST IS SORTED.



TASK 4 – SQL TO PANDAS QUERY

PROMPT :
CONVERT SQL QUERY INTO PANDAS CODE.

EXPLANATION :

THE SQL FILTER CONDITION IS APPLIED USING PANDAS. IT SELECTS ROWS WHERE SALARY IS GREATER THAN 50000 AND SHOWS REQUIRED COLUMNS.

Welcome | # Write a Python function to fetch curre.py

Users > ronithreddyrevuri > Documents > AI Asstd Coding_COMPL > # Write a Python function to fetch curre.py > ...

```python
# SQL to Pandas code

import pandas as pd

data = pd.DataFrame({
    "name": ["A", "B", "C"],
    "salary": [40000, 60000, 70000]
})

result = data[data["salary"] > 50000][["name", "salary"]]

print(result)
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | AZURE | QUERY RESULTS

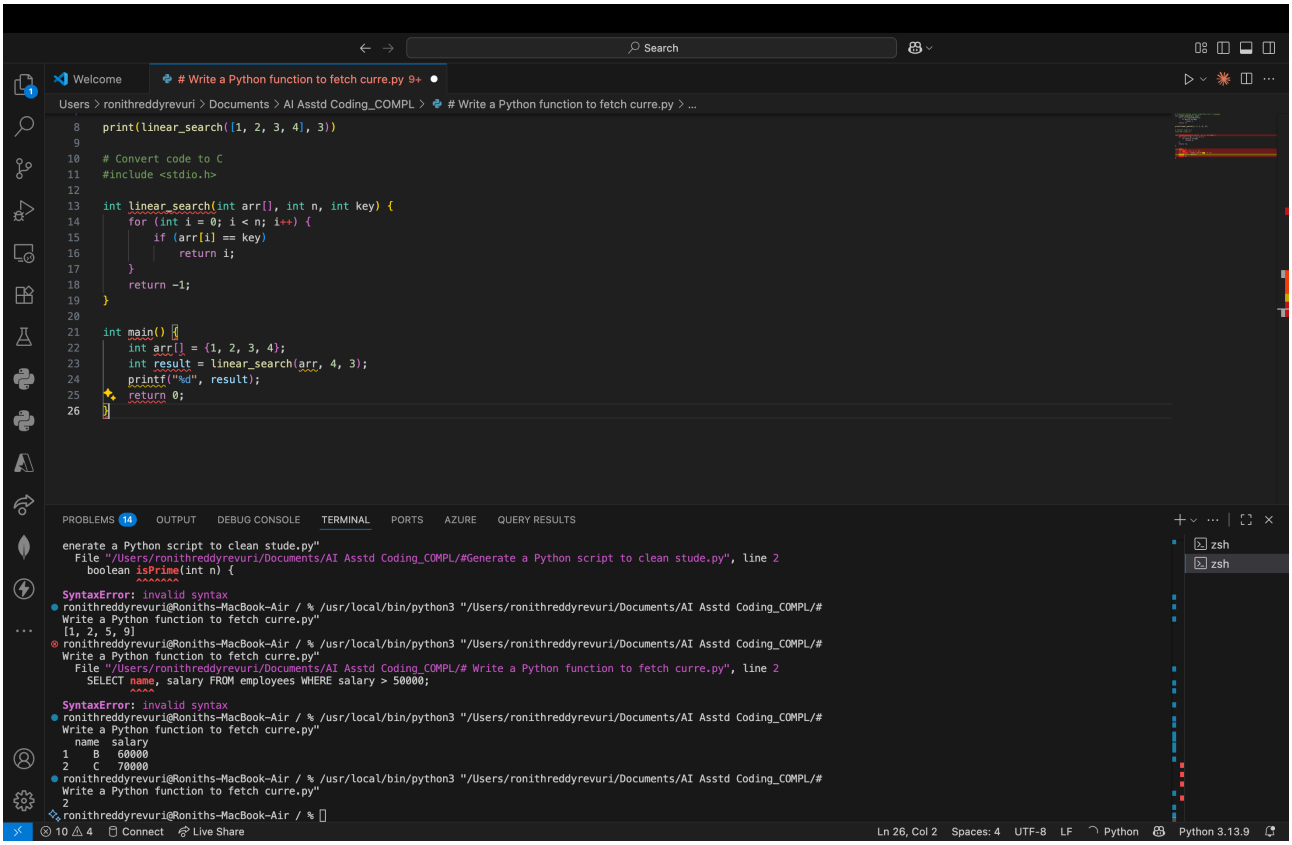
```
                ^
SyntaxError: invalid syntax
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/#G
enerate a Python script to clean stude.py"
  File "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/#Generate a Python script to clean stude.py", line 2
    boolean isPrime(int n) {
            ^^^^^^

SyntaxError: invalid syntax
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/#
Write a Python function to fetch curre.py"
[1, 2, 5, 9]
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/#
Write a Python function to fetch curre.py"
  File "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/# Write a Python function to fetch curre.py", line 2
    SELECT name, salary FROM employees WHERE salary > 50000;
           ^^^^

SyntaxError: invalid syntax
ronithreddyrevuri@Roniths-MacBook-Air / % /usr/local/bin/python3 "/Users/ronithreddyrevuri/Documents/AI Asstd Coding_COMPL/#
Write a Python function to fetch curre.py"
  name  salary
1    B   60000
2    C   70000
ronithreddyrevuri@Roniths-MacBook-Air / %
```

zsh
zsh

0 0 | Connect | Live Share | Ln 1, Col 1 | Spaces: 4 | UTF-8 | LF | Python | Python 3.13.9

TASK 5 – REAL-TIME APPLICATION: ALGORITHM TRANSLATION ACROSS LANGUAGES

PROMPT :
TRANSLATE PYTHON SEARCH ALGORITHM INTO C LANGUAGE.

EXPLANATION

THE PYTHON SEARCH FUNCTION IS CONVERTED INTO C USING ARRAYS AND LOOPS. SYNTAX CHANGES BUT LOGIC REMAINS THE SAME.